

Read the following case study and answer ALL questions.

Johor Putera Specialist Hospital (JPSH) is a subsidiary of Johor Healthcare Berhad (KPJHB), the largest group of private healthcare provider in Malaysia. Currently, more than 10,000 patients are registered with a JPSH. The JPSH employs a number of general practitioners (i.e. doctors, specialist) and a few receptionists.

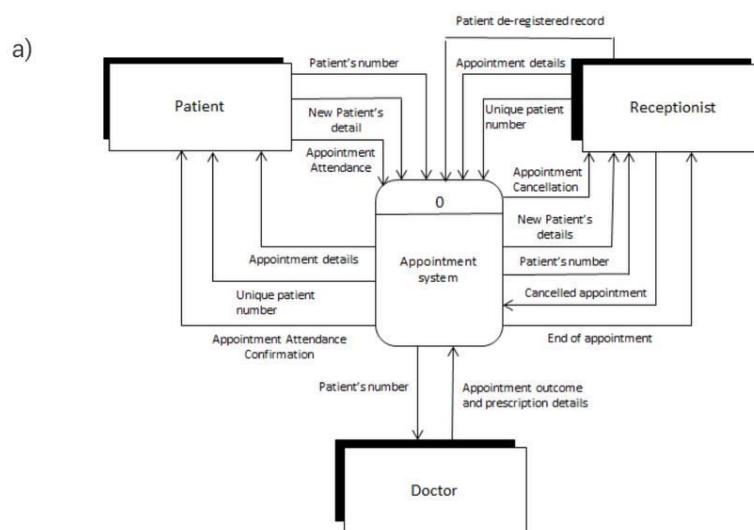
Patients are officially registered with one doctor but can arrange appointments with any available one. These appointments may subsequently be cancelled. Some appointments result in one or more prescriptions, identifying a medicine to be taken. New patients are registered by a receptionist. When a patient is registered he/she provides his/her details such as name, date of birth, address, etc., and receives a unique patient number.

To book an appointment a patient should contact a receptionist. The patient provides his/her number (or date of birth) and the receptionist provides a list of available time slots for appointments. The appointment is booked with the patient's doctor or if the patient's doctor is not available with any available doctor. The date and time of the booked appointment are given to the patient as a confirmation.

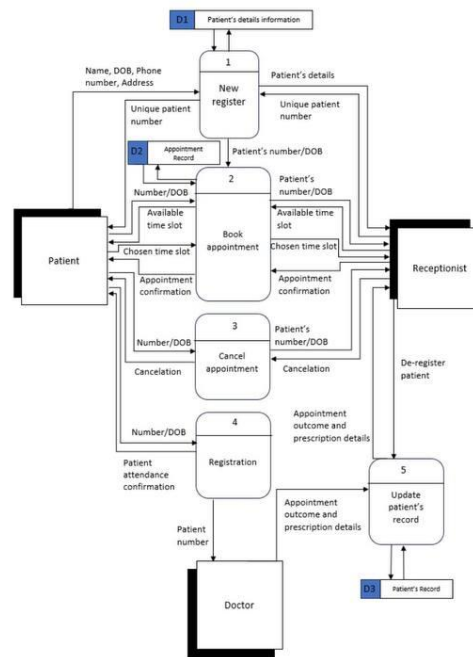
Patients can cancel booked appointments by contacting a receptionist who will cancel appointments on behalf of patients. A patient who attends an appointment should check in first using a special terminal located in the waiting area. The patient inputs his/her number (or date of birth). The system checks the details and confirms that the patient has been checked in.

Doctors record appointment outcomes and details of prescriptions (if any) during the appointments i.e. all prescriptions issued by doctors are recorded on the patient's record. Patients who leave the area where the JPSH is located are de-registered by a receptionist.

a) Develop a context diagram of the JPSH system described in the case study.



b) Develop the Diagram 0 of the JPSH system.



c) One of the processes identified in question (b) is **BOOK AN APPOINTMENT**. Develop a Level 1 Diagram for the process.

